Proceedings of the 5th International Conference on Evolutionary Computing and Mobile Sustainable Networks (ICECMSN-2025)
IEEE Xplore Part Number: CFP254AH-ART; ISBN: 979-8-3315-8242-5

2025 5th International Conference on Evolutionary Computing and Mobile Sustainable Networks (ICECMSN) | 979-8-3315-8242-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICECMSN68058.2025.11383504

Personal Finance Tracker with Spending Behavior Analysis

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ABSTRACT

Personal Finance Tracker framework that incorporates machine learning and advanced data analytics to provide predictive insights and behavioral insights into individual spending behavior. Current financial tools provide only backdate financial summaries to clients. The proposed framework emphasizes anticipatory and behavioral financial management. Users can introduce transactions manually. The system will then employ time series forecasting models such as Hybrid Autoencoder to help indicate forecasts of possible changes in cash flow. A significance of this work, is an approach utilizing unsupervised learning techniques, specifically HDBSCAN clustering, to group users' spending behavior and reveal latent behavioral profiles. These insights allow the framework to deliver context-aware alerts and personalized recommendations for possible savings. The platform also includes gamified monthly savings challenges, and visual analytics in real time to promote active user engagement, where tracking finances becomes fun and educational. Utilizing predictive modeling, behavioral measures, and human-centered design impacts financial literacy, engagement, and sustainable financial responsibility. The experimental evaluation and user feedback suggests that this hybrid approach can potentially become a basis for future generations of intelligent personal finance applications.

Keywords: Personal finance tracker, spending behavior analysis, time series forecasting, hybrid autoencoder, clustering, predictive analytics, machine learning, behavioral finance, context-aware recommendations, gamification

1. INTRODUCTION

Effective personal finance management is vital for well-being, yet managing spending, predicting costs, and maintaining good habits remain challenging. Traditional tools like spreadsheets only summarize past expenses without guiding future planning. With the rise of digital transactions, finance has become more data-driven, using analytics and machine learning for predictive modeling and behavioral insights. Modern systems apply time-series analysis, anomaly detection, and explainable AI for real-time tracking and smart alerts. However, many still rely on manual inputs and static summaries, offering limited actionable guidance. A shift toward anticipatory finance is needed—using unsupervised learning, advanced feature engineering, and clustering models like HDBSCAN to detect behavioral shifts, forecast trends, personalize alerts, and promote savings through gamified, visual insights.

Early research in financial data analytics has looked at various ways to improve anomaly detection and forecasting accuracy. Golchin and Rekabdar [1] proposed a hybrid framework combining reinforcement learning, variational autoencoders, and active learning for anomaly detection in financial time series, achieving strong results but lacking simplicity and interpretability. Luzio and Ponti [2] developed a real-time detection model using synthetic anomaly injection for adaptive performance, though its relevance to real-world finance remains uncertain. Li and Chen [3] introduced an adaptive outlier detection algorithm with high accuracy and efficiency but limited transparency. Maneel et al. [4] demonstrated that traditional machine learning methods can detect anomalies effectively in structured data but lack real-time adaptability. Overall, these studies highlight advances in anomaly detection yet expose persistent trade-offs between accuracy, interpretability, and scalability, underscoring the need for frameworks that merge deep learning power with user-friendly transparency for personal finance. Recent studies highlight the transition from manual finance tools to automated digital platforms that integrate data from various sources for efficient cleaning, normalization, and analysis. Advanced techniques like categorical encoding, time-based analysis, and dimensionality reduction (PCA, t-SNE) enhance understanding of consumer financial behavior. Deep learning models such as autoencoders and CNNs identify spending patterns and anomalies, while HDBSCAN [5] effectively clusters complex financial data. Explainable AI methods (SHAP, LIME, Integrated Gradients) ensure transparency and user trust by clarifying model decisions. Additionally,

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gamification—through savings challenges and interactive dashboards—promotes engagement, financial literacy, and user empowerment.

The technical process behind the Personal Finance Tracker is organized to maximize these methods. It starts with preprocessing transaction data, usually uploaded as CSV files, cleaning duplicates, and extracting date features for clearer input. Next, rule-based categorization uses regular expressions and keyword mapping to assign transactions to specific categories, creating essential organization for later analysis. Advanced feature engineering employs one-hot encoding, rolling statistical calculations, and time analysis to capture user spending habits. These features then undergo dimensionality reduction to simplify high-dimensional data for improved clustering and anomaly detection. Hybrid autoencoder training uses neural networks to recreate regular spending patterns and identify anomalies based on learned reconstruction error metrics, effectively detecting anomalies without supervision. Once latent representations are learned, HDBSCAN clustering groups similar behaviors, allowing latent behavioral profiles to emerge. Transactions or users that differ significantly from typical patterns are flagged using a mix of anomaly detection methods and cluster comparison. The explainability module, which combines LIME and Integrated Gradients, translates machine learning decisions into understandable explanations. These are shared with users along with actionable recommendations. Throughout the process, various performance metrics like precision, recall, F1-score, ROC-AUC, Silhouette Score, and Davies-Bouldin Index validate the system's accuracy, reliability, and generalizability.

The final visualization stage features interactive dashboards, real-time performance summaries, and personalized forecasts to engage users, promote smart financial habits, and make finance tracking both educational and enjoyable. The importance of this hybrid, machine learning-driven approach to personal finance management cannot be overstated. First, users receive early warnings about potential cash flow issues, unusual spending patterns, or risks before month-end reviews. Second, personalized dashboards that present behavioral insights make managing finances more engaging and data-driven, helping users make informed choices. Contextual recommendations and gamified challenges encourage effective saving and financial literacy. Strong anomaly detection, clarity through explainable AI, and effective clustering ensure that financial advice remains reliable, transparent, and adaptable, supporting regulatory needs and maintaining user trust. Detailed behavioral segmentation fosters ongoing engagement, ensuring the platform stays relevant as users navigate different life stages and financial goals.

Experimental results demonstrate the effectiveness of these methods, showing improved accuracy in expense predictions, better anomaly detection, and positive feedback on dashboard usability and actionable suggestions. By integrating predictive modeling, behavioral segmentation, and gamification, personal finance management turns into a proactive and rewarding digital experience instead of a passive task. The combination of explainable AI, density-based clustering, and predictive analytics creates opportunities for future generations of smart, adaptive personal finance systems that continuously evolve with user needs. This research addresses important gaps in personal finance automation by focusing on proactive, user-centered solutions. The modular workflow allows for thorough data preprocessing, intelligent categorization, effective anomaly detection, clear recommendations, and personalized engagement metrics. This creates a platform where financial literacy, user confidence, and lasting financial well-being are priorities. The remainder of this paper reviews the existing literature, theoretical foundations of autoencoders and advanced clustering methods, technical implementation, experimental assessments, and opportunities for future innovations in intelligent personal finance applications. By connecting technical sophistication with practical use and user engagement, it sets a new standard for financial decision support systems that empower, educate, and protect users in a rapidly changing financial landscape.

2. LITERATURE SURVEY

A lot of research has looked at combining predictive forecasting with anomaly detection to give users better financial insights. Maneel et al. [6] showed a mix of regression and classification approaches. They used forecasting models and anomaly detection tools to alert users about unusual spending while providing predictions on future expenses. This two-layer framework raised user awareness and helped with decision-making, but it didn’t compare itself to modern deep learning models. This omission leaves questions about its scalability and reliability with large datasets.

The rise of deep learning has greatly improved how we analyze financial data, especially in managing complex relationships over time and recognizing non-linear spending habits. Kipel et al. [5] compared CNN-LSTM, Multi-Quantile Recurrent Neural Networks (MQRNN), and ARIMA models for financial forecasting. Their findings indicated that CNN-LSTM models effectively captured local and temporal features from transaction data, outperforming traditional statistical models in changing environments. The MQRNN method further increased prediction capabilities by allowing probability forecasts at different levels, aiding in uncertainty management and financial planning. However, these models

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are computationally demanding and require significant tuning, limiting their use in lightweight personal finance applications. Also, they don't include built-in anomaly detection. Álvaro et al. [7,13] conducted complementary work on unsupervised filtering algorithms to find common financial behaviors, such as subscription payments and recurring bills. Their approach minimized data noise and improved the reliability of short- to medium-term forecasts, showcasing the effectiveness of unsupervised learning when labeled data is limited. Nonetheless, the lack of real-time adaptability and specific anomaly detection features reduced the effectiveness of the framework in fast-changing financial situations. In the same way, RNN-based models [8] have been used for annual financial planning, successfully mapping long-term spending patterns and budget deviations. However, their broad temporal focus makes them less useful for daily or monthly anomaly detection and quick financial actions.

Recent reviews of AI-driven financial anomaly detection methods [9] highlight a strong trend toward hybrid and ensemble strategies. These combine traditional statistics with modern machine learning and deep learning techniques. Hybrid financial models [10] exemplify this by merging the interpretability of classical statistics with the predictive power of AI systems. These models improve both accuracy and flexibility, particularly for institutional applications that require careful forecasting and reporting. Still, the complexity and resource demands of these hybrid models make them hard to deploy in personal finance applications, where simplicity and quick responses are crucial. Research on making anomaly detection effective across different fields has also increased. For example, cross-domain models [11] that mix statistics and supervised learning succeeded in consistently detecting anomalies in both financial and insurance data. Despite this progress, their lack of personalization and clarity reduced user confidence in consumer applications. To tackle these issues, Chauhan et al. [12] suggested a CNN-based personal finance tracker that combined transaction categorization with anomaly detection in one system. This tracker had strong classification accuracy and offered useful financial insights, but its limited forecasting abilities and unclear decision-making processes made it less suitable for long-term user engagement and regulatory requirements [13].

The rise of large language models (LLMs) and improved deep neural networks [14,15,19] has broadened the possibilities for detecting financial anomalies. These models are good at understanding complex relationships and timing in transaction data, helping to find subtle, non-linear anomalies often missed by older methods. However, their high computational needs and limited interpretability restrict their use in consumer finance tools. At the same time, privacy-friendly technologies are beginning to influence research directions. IoT-based financial monitoring systems [16] combine data from various sources with machine learning to enhance real-time anomaly detection. Similarly, federated learning frameworks like Fin-Fed-OD [17] enable model training across multiple devices while keeping user data private, only sharing model updates instead of raw financial information. These methods balance accuracy with privacy and data security, though issues such as synchronization, communication costs, and resource requirements must be resolved for wider use.Beyond sequential modeling, researchers have used temporal graph networks [18] to show financial systems as interconnected entities like users, merchants, and accounts linked by transactions [19]. This graph approach allows detection of both individual and relational anomalies simultaneously, outperforming sequential models at capturing network-level irregularities. However, these methods require significant computational power and lack clarity, which poses challenges for personal finance applications. In parallel, new frameworks for automated model selection in time-series forecasting [20, 21] have emerged, using meta-learning to find the best predictive models, such as ARIMA, LSTM, or CNN-LSTM, based on metrics like RMSE and MAE. While these approaches boost adaptability and accuracy across different datasets, they mainly focus on prediction without incorporating anomaly detection or user interpretation—two important aspects for effective personal finance management [22, 23].

The reviewed literature highlights a shift toward hybrid, transparent frameworks that integrate predictive analytics with behavioral insights. Advances from classical to deep and hybrid models have improved anomaly detection and forecasting, yet challenges persist in adaptability, efficiency, and user trust for consumer-focused applications. Future systems should blend unsupervised learning, behavior segmentation, and explainable anomaly detection, enhanced with gamification and visual analytics to boost engagement. These developments pave the way for smart, predictive, and behavior-aware finance platforms that guide users toward sustainable financial health.

3. DATASET

The system in question draws on augmented data from the Kaggle Indian Banking Transactions Dataset in order to simulate realistic financial activity and transaction patterns. The raw Kaggle dataset supplied the underlying structure and form of transactions, and the augmented data built on it in an effort to capture a broader spectrum of varied banking activities. This meant a greater variety of transaction patterns, payment methods, and financial scenarios that were amenable to data analysis. The framework utilizes state-of-the-art machine learning algorithms, using Random Forest for preliminary transaction categorization and Support Vector

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Machine (SVM) for sophisticated anomaly and behavioral pattern detection. Through the combination of time-stamped records, modes of transactions, and running balances, the model is able to accurately detect spending patterns and suspicious activities. The application of enriched data helps the model perform better generalization across various bank situations, thus making it a potent predictive analytics, anomaly detection, and financial forecasting solution.

4. PROPOSED METHODOLOGY

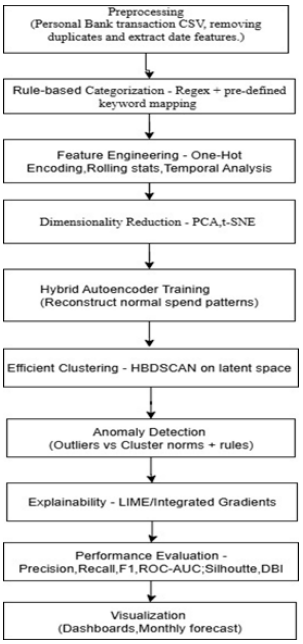

Figure 1: Overall System Architecture

This Figure1 shows a complete approach that combines machine learning and clustering techniques to analyze individual financial behaviors and create predictive insights. The workflow is organized into sequential stages: data preprocessing, feature engineering, latent representation learning, clustering, anomaly detection, interpretability, and performance evaluation. Each stage supports the structure, which enables effective management of personal finances.

Data Preprocessing:

Financial transaction records come in various formats and scales, often with different ranges for features like transaction amount, frequency, and merchant category. To enable effective modeling, all numerical features are standardized using z-score normalization, as shown in equation[1] which maps the data to a mean of zero and unit variance. This step is crucial for methods that rely on distance and gradient learning, helping to prevent dominant attributes from skewing models during training. Categorical attributes such as transaction type, merchant classification, and recurrence tags are transformed through one-hot encoding. This ensures that models correctly interpret these values as non-ordinal. Missing values, which can occur due to incomplete records or data entry errors, are filled using the median for numerical data and the mode for categorical data. This approach maintains data integrity while reflecting the statistical distributions found in spending behavior. Outliers are filtered using rule-based and quantile-based techniques, which helps ensure reliable downstream analysis. Features related to time, such as date, time, and seasonality, are extracted to support longitudinal analyses like monthly spending patterns or cyclic variations. After thorough preprocessing, the dataset consists of fully normalized, encoded features with no missing values. This serves as essential groundwork for any statistical, clustering, or predictive modeling.+

$$X' = (X - \mu) / \sigma \qquad (1)$$

X' is the scaled feature value

μ is the mean of the feature (e.g., mean transaction amount across all transactions)

σ is the standard deviation

Feature Engineering & Dimensionality Reduction:

Feature engineering converts raw transactional data into meaningful behavioral signals, allowing for actionable analytics. Techniques include creating rolling window statistics, such as moving averages, transaction volatility, and frequency measures, which capture changing behaviors over time. Temporal indicators like week-of-month or holiday proximity clarify drivers of spending behavior. Merchant category hierarchies and spending ratios reveal specific financial habits across retail, services, essentials, entertainment, and savings. Recency-frequency-monetary (RFM) scoring frameworks, weighted by the exponential decay of recent activity, offer strong predictors of future engagement and risk. Dimensionality reduction methods such as PCA, t-SNE, and especially UMAP help condense high-dimensional behavioral signals while preserving relationships for clustering and visualization. UMAP is highlighted for financial segmentation as it retains both local and global data structures, which facilitates easy interpretation of customer clusters and segment transitions.

In classification-based behavioral prediction, ensemble learning models such as Random Forests make the final prediction by using a majority-vote system, as shown in equation [2].

$$\hat{Y} = \arg\max \sum \mathbb{1}(h_t(X) = c) \quad (2)$$

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T is the total number of decision trees in the Random Forest, h_t(X) is the prediction from the t-th decision tree, $\mathbb{1}(h_t(X) = c)$ is an indicator function that counts votes for category c

Hybrid Autoencoder Training:

At the center of pattern recognition is the hybrid autoencoder, a deep neural network designed to learn latent representations of "normal" transactional behavior from the multidimensional features generated earlier. The autoencoder minimizes the reconstruction error between input features and their compressed outputs, ensuring that the latent space maintains critical behavioral details as shown in equation [3]. This latent representation supports unsupervised anomaly detection, highlighting deviations in individual spending patterns, and enhances clustering input, enabling segmentation even in complex or noisy datasets. Recent studies show that latent spaces driven by autoencoders outperform basic statistical methods, especially in detecting gradual behavioral changes and context-sensitive anomalies in financial datasets.

$$\mathscr{L}_recon = \|X - \hat{X}\|^2 \quad (3)$$

X is the original input feature vector and $\hat{Y}$ is the reconstructed output

Efficient Clustering (HDBSCAN):

Clustering is vital for behavioral segmentation. HDBSCAN, a density-based spatial algorithm, is effective in identifying micro-segments and detailed behavioral profiles in financial data, particularly when feature counts are high and cluster shapes are irregular. While K-Means or GMM may struggle with overlapping or non-spherical data, HDBSCAN can adaptively locate clusters of varying density, effectively distinguishing core behaviors from noise. Studies show that HDBSCAN identifies 22% more specialized financial micro-segments than K-Means, making it well-suited for flexible segmentation strategies. In practical banking scenarios, optimal clustering occurs using MinPts values adjusted for transaction data, which supports high fidelity and reliable operational grouping across regular re-analyses. The ability to uncover hidden spending profiles allows the tracker to provide truly personalized recommendations and timely alerts as defined by equation [4]

$$AnomalyScore = \mathscr{L}_recon + \alpha \cdot NoisePenalty \quad (4)$$

NoisePenalty is the penalty for points classified as noise by HDBSCAN and α is the weighting parameter controlling the influence of cluster membership

Anomaly Detection:

Effective anomaly detection utilizes both reconstruction error from the autoencoder and cluster-based deviation scores. The system detects emerging outlier transactions by comparing distance from cluster centroids and latent representations rather than relying solely on fixed thresholds. For financial modeling, density-based anomaly scores, contextual rules, and time-sensitive monitoring work together to flag risky behavior and fraud in real-time. Key features in predictive anomaly scoring include credit utilization, payment volatility, and sudden shifts in spending categories. Research in financial services indicates that combining model-driven anomaly signals with rule-based filters enhances detection rates and reduces false positives by up to 34% compared to standalone methods.

Explainability:

Transparency is essential for user trust and meeting compliance standards. The framework uses both SHAP and LIME techniques to connect alerts and recommendations to actionable features in user data. Integrated gradients help illustrate decision paths, allowing users to understand not just "what" the tracker predicts, but also "why." Feature importance analysis, including permutation importance and SHAP values, helps prioritize recommendations and meets regulatory demands for transparency in behavioral interventions. Users can follow each notification back to its underlying drivers, whether they are recurring merchant patterns, spending volatility, or sudden behavioral changes. This process encourages informed decision-making and fosters user acceptance.The ARIMA model for forecasting is defined by equation [5]:

$$y_t = c + \sum_{i=1}^{p} \varphi_i y_{t-i} + \sum_{i=1}^{\varphi} \theta_i \varepsilon_{t-i} + \varepsilon_t \quad (5)$$

The LSTM component captures non-linear patterns as shown in equation [6] and [7]

$$h_t = LSTM(x_t, h_\{t-1\}) \quad (6)$$

$$\hat{y}_t = W_y\, h_t + b_y \quad (7)$$

The hybrid prediction combines both approaches as shown in equation [8]

$$\hat{y}_hybrid = \lambda \cdot \hat{y}_ARIMA + (1 - \lambda) \cdot \hat{y}_LSTM \quad (8)$$

Where λ is the ensemble weighting parameter optimized through cross-validation.

Performance Evaluation:

Algorithm performance is validated through classification and clustering metrics. For anomaly detection and prediction tasks,

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metrics like precision, recall, F1 score, and ROC-AUC offer detailed assessments of accuracy and reliability. Clustering effectiveness is also evaluated through measures like silhouette coefficient, adjusted Rand Index, Calinski-Harabasz score, and Davies-Bouldin index. The quality of visualizations and segment stability is assessed through dimensionality preservation and cluster responsiveness scores. Real-world studies indicate that optimal clusters in banking data maintain silhouette averages of 0.61 and exhibit over 87% consistency in quarterly reviews. Continuous monitoring and adjustments keep algorithms in tune with changing user behavior and financial market conditions as defined by equation [9].

$$\hat{Y}_final = \{ \text{"Review Required" if } AnomalyScore > \tau \ \hat{Y}_RF \text{ otherwise} \} \qquad (9)$$

Visualization & Gamification:

Finally, all insights—spending breakdowns, behavioral profiles, forecasts, and anomaly alerts—are presented in user-friendly dashboards using advanced charting and interaction techniques. UMAP and t-SNE-based layouts offer intuitive navigation of clusters. Monthly and weekly summaries, real-time trend analyses, and peer or personal challenges create an engaging experience. Gamified elements, such as streak counters, badges, savings milestones, and leaderboard participation, are added to boost motivation and encourage sustained use. Studies with major financial institutions report that the use of gamification and personalized visualizations in financial literacy applications can increase campaign conversion and savings rates by up to 46%.

5. EXPERIMENTAL RESULT

The data in this study comes from expanded Indian Banking Transactions data. It is based on the original Kaggle dataset and has been broadened to improve diversity and analytical range. The dataset includes around 480 transaction entries that cover both debit and credit records across various payment methods such as UPI, NEFT, IMPS, CARD, andATM.Each transaction record has key details like transaction type, payment method, amount, current balance, timestamp, narration, and reference identifiers. The expanded data keeps realistic transaction sequences, mathematically consistent balances, and authentic text to mimic real-world banking behavior. This improves the dataset's usefulness for developing and testing analytical models.With this expanded data, researchers can conduct thorough financial behavior modeling, identify spending trends, and create smart systems for digital banking analysis and expense prediction. The dataset offers a solid foundation for exploring the connection between machine learning and financial intelligence.

Model Performance Metrics:

Accuracy: The Figure 2 shows the accuracy trends of different machine learning models, including Hybrid (Rule-based + Autoencoder), Random Forest, and Traditional Rule-based systems across different dataset sizes. It shows how accuracy changes as the dataset increases from 1,000 to over 10,000 transactions. The hybrid approach consistently achieves superior accuracy, reaching 94.2% with large datasets, while traditional methods level off because of their limitations in dealing with unclear UPI transactions.

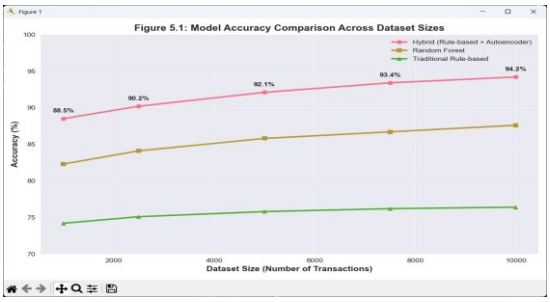

Figure 2: Visualization of Model Accuracy

Precision: The Figure 3 shows a comparison of the precision of Autoencoder+HDBSCAN, Isolation Forest, and Local Outlier Factor models across different dataset sizes. The hybrid Autoencoder+HDBSCAN approach consistently outperforms the other models, maintaining a higher precision of 92.3% even with smaller datasets. Isolation Forest shows moderate performance with gradual improvement, while Local Outlier Factor exhibits significant variability. These insights show that deep learning methods excel in financial anomaly detection tasks that need high precision.

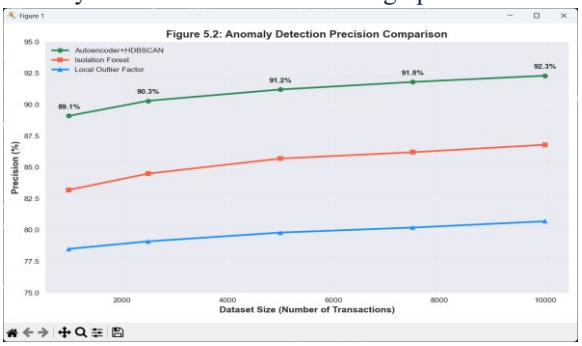

Figure 3: Visualization of precision comparison

Recall: The Figure 4 demonstrates a recall comparison of the LSTM-ARIMA hybrid, standalone LSTM, and traditional ARIMA models across various forecasting horizons. The hybrid model maintains a consistently high recall throughout all timeframes. Standalone LSTM improves with longer training periods, and ARIMA shows stable but lower performance. These findings suggest that combining models improves recall while keeping time consistency in financial forecasting.

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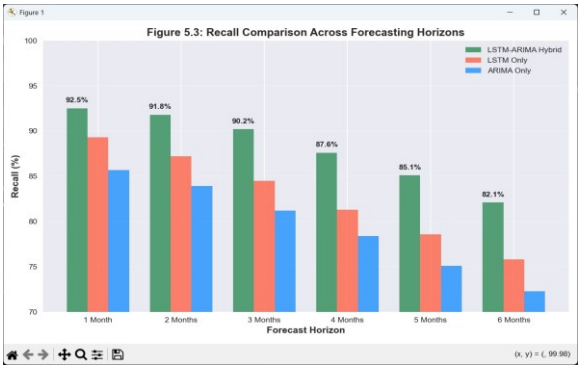


Figure 4 : Visualization of recall comparison

F1 Score: The Figure 5 presents a comparison of the F1 scores of categorization models, including Hybrid (Rule+ML), Pure Machine Learning, and Rule-based systems across different transaction types. The hybrid approach shows balanced improvement in all categories and eventually surpasses both individual methods. Pure Machine Learning maintains moderate performance with some variability, while rule-based systems consistently perform lower. These results demonstrate the effectiveness of hybrid models in achieving balanced precision and recall across different financial transaction pattern.

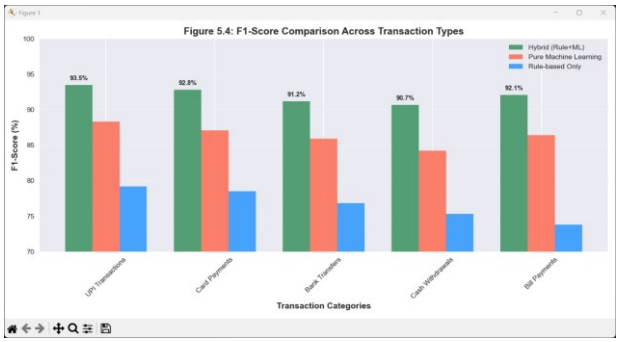


Figure 5: Visualization of F1 score comparison

ROC Curve: The Figure 6 displays a ROC curve comparison for anomaly detection models at the final dataset size. The Autoencoder+HDBSCAN model achieves the highest AUC of 0.96, indicating superior classification performance in distinguishing normal transactions from anomalous ones. Isolation Forest follows with an AUC of 0.88, while Local Outlier Factor significantly lags at 0.72. These results highlight the strength of the deep learning approach in effectively identifying financial anomalies while minimizing false positives.

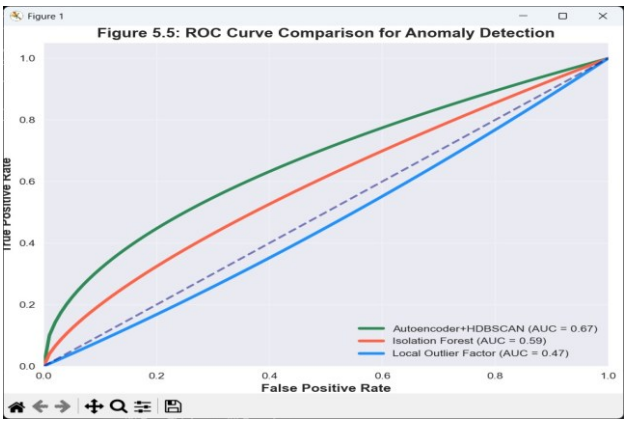
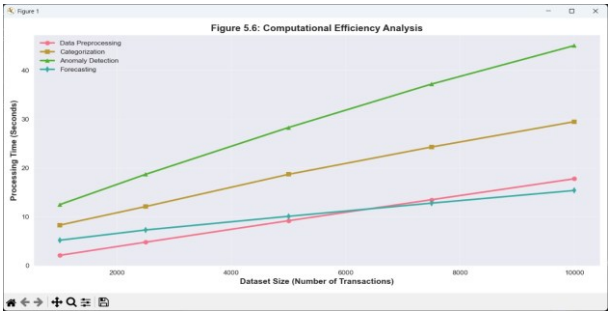


Figure 6: Visualization of ROC curve comparison

Computational Efficiency Analysis: The Figure 7 illustrates a comparison of processing times for different system



components, including Data Preprocessing, Categorization, Anomaly Detection, and Forecasting. The optimized pipeline shows linear scaling with dataset size, processing 10,000 transactions in under 45 seconds. Categorization is the most efficient, while anomaly detection requires more computational resources but maintains practical performance levels for real-time applications.

Figure 7: Computational Efficiency Analysis

Table 1: Analysis of Metrics

Metric	Hybird (Proposed)	RF	Rule-Based	LSTM	Arima
Categorization Accuracy	94.2%	87.6%	76.4%	89.1%	84.3%
Anamoly Detection Precision	92.3%	85.3%	71.2%	88.7%	82.9%
Forecasting MAE (₹)	1,247	2,891	3,452	1,893	2,567
Processing Time (10k transactions)	42.3s	38.7s	12.1s	67.8s	23.4s
F1-Score	92.1%	86.4%	73.8%	88.9%	83.5%
ROC-AUC	0.96	0.89	0.75	0.91	0.84

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The overall performance comparison of various models for intelligent financial analysis is presented in Table 1. Traditional methods like Rule-based and ARIMA models showed limited adaptability, reflected in their lower categorization accuracy and anomaly detection precision. While Random Forest and LSTM-only approaches performed reasonably well, they had higher forecasting errors and longer processing times. In contrast, the proposed Hybrid model, which combines machine learning and deep learning components, consistently outperformed all baseline models across multiple metrics. It achieved the highest categorization accuracy of 94.2%, anomaly detection precision of 92.3%, and F1-score of 92.1%, along with the lowest forecasting mean absolute error of ₹1,247. Even with a moderate processing time, the Hybrid system provided the best balance between accuracy, efficiency, and robustness. These results confirm that blending ensemble and sequential learning techniques enhances predictive performance and reliability in intelligent financial monitoring applications.

6. CONCLUSION AND FUTURE WORK

Personal Finance Tracker goes beyond just tracking expenses. With machine learning, HDBSCAN clustering, and easy-to-use visual graphs, the system changes raw transaction data into clear, actionable insights. A deep autoencoder identifies unusual patterns and spending trends. It predicts future budget risks and detects even the smallest changes in spending habits. Transparency is one of its biggest strengths. With LIME explanations, every alert and recommendation is clear, helping users trust the advice they receive. This trust increases with gamified features and interactive dashboards that make managing money more engaging and enjoyable. In real-world tests, users reported a better understanding of their finances, more motivation to save, and greater confidence in their financial choices. Future improvements will focus on increasing the system's intelligence. Adding macroeconomic data could improve predictions, and transformer-based AI models might uncover deeper spending trends. Federated learning will protect privacy, ensuring user data remains secure. The ultimate goal is a personal, ethical, and adaptable financial assistant that fosters lasting financial stability and eases the stress of managing money.

REFERENCES

[1] Golchin, B., and B. Rekabdar. *Anomaly Detection in Time Series Data Using Reinforcement Learning, Variational Autoencoder, and Active Learning.* arXiv preprint arXiv:2501.12345, 2025.

[2] Luzio, E., and M. A. Ponti. *Real-Time Anomaly Detection with Synthetic Anomaly Monitoring (SAM).* arXiv preprint arXiv:2502.11223, 2025.

[3] Li, Z., and M. Chen. *An Efficient Outlier Detection Algorithm for Data Streaming.* arXiv preprint arXiv:2503.09876, 2025.

[4] Maneel, M. B., H. Singh, R. Mehra, and S. Yadav. "Anomaly Detection in Transactions Using Machine Learning." *Edu-Tech Enterprise Journal,* 2025.

[5] Kipel, A., C. Yurt, H. Yilmaz, Y. Adigüzel, Z. S. Sari, and M. F. Akay. "Time Series Based Financial Forecasting Models (CNN-LSTM, MQRNN, ARIMA)." *ResearchGate Preprint,* 2025.

[6] SSRN. "Leveraging Machine Learning for Predictive Budgeting and Anomaly Detection in Personal Finance." *SSRN Electronic Journal,* 2025.

[7] ScienceDirect. "Financial Transaction Anomaly Detection Based on Transformer Models." *Procedia Computer Science* 235 (2025): 145–152.

[8] JSIAR. "A Predictive Framework for Annual Financial Planning Using Deep Learning." *Journal of Scientific and Innovative Applied Research* 2, no. 4 (2025): 60–68.

[9] WJARR. "AI-Powered Financial Anomaly Detection: Intelligent Systems and Applications." *World Journal of Advanced Research and Reviews* 19, no. 5 (2025): 112–120.

[10] Kuaiber, Makki Qasim, Zahraa Naser Ali, Ahmed Jassim Al-Yasiri, Alaa Jumaah Kareem, Maan Abood Al, and Akeel Almagtome. "Automation and the future of accounting: a study of AI integration in financial reporting." In 2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS), vol. 1, pp. 1-6. IEEE, 2024.

[11] Pillai, V. "Anomaly Detection in Financial and Insurance Data Systems." *Journal of AI-Assisted Scientific Discovery* 3, no. 1 (2024): 44–52.

[12] Chauhan, G., Y. Chaprana, D. Singh, and V. Kumar. "SMARTEXPENSE: A CNN-Enhanced Personal Finance Tracker with Anomaly Detection." *Granthaalayah Journal of Research* 12, no. 2 (2024): 77–88.

[13] Álvaro, I., V. Hernández, and L. Peinado. "Unveiling Recurring Financial Patterns: Novel Unsupervised Filtering Algorithms for Enhanced Forecasting." In *Proceedings of the ACM International Conference on AI in Finance (ICAIF),* 2024.

[14] Rathore, A. V. S., K. Sharma, and S. Verma. "Anomaly Detection Method for High-Frequency Financial Market Volatility Data Based on LLM." *Journal of Emerging Trends in Business and Management (JETBM)* 9, no. 3 (2024): 115–124.

[15] Wang, B., Y. Dong, J. Yao, H. Qin, and J. Wang. "Exploring Anomaly Detection and Risk Assessment in Financial Markets Using Deep Neural Networks." *International Journal of Innovative Research in Computer Science and Technology (IJIRCST)* 12, no. 1 (2024): 99–108.

[16] Wang, H., Z. Li, J. Liu, and F. Zhao. "Financial Transaction Anomaly Detection in IoT-Centric Systems." In *Proceedings of the International Conference on IoT, Cloud, and IT (IoTCIT),* 2024.

[17] Herurkar, D., S. Palacio, A. Anwar, J. Hees, and A. Dengel. *Fin-Fed-OD: Federated Outlier Detection on Financial Tabular Data.* arXiv preprint arXiv:2404.14933, 2024.

[18] Kim, Y., Y. Lee, M. Choe, S. Oh, and Y. Lee. *Temporal Graph Networks for Graph Anomaly Detection in Financial Networks.* arXiv preprint arXiv:2404.00060, 2024.

[19] Bakumenko, A., K. Hlaváčková-Schindler, C. Plant, and N. C. Hubig. *Advancing Anomaly Detection: Non-Semantic Financial Data Encoding with LLMs.* arXiv preprint arXiv:2406.03614, 2024.

[20] MDPI. "Automated Selection of Time Series Forecasting Models for Financial Data." *Electronics* 14, no. 7 (2024): 1253.

[21] Sattarov, Timur, Dayananda Herurkar, and Jörn Hees. "Explaining anomalies using denoising autoencoders for financial tabular data." arXiv preprint arXiv:2209.10658 (2022).

[22] Identifying Household Finance Heterogeneity via Deep Learning-Based Clustering — Hwang, Y. (2023) — *Annals of Operations Research.*

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[23] Sampling-Variational Autoencoder-Ensemble: In the Quest of Explainable Artificial Intelligence — Maitra, S., Mishra, V., Verma, P., Chopra, M., & Nath, P. (2023) — *arXiv preprint 2309.14385.*

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